

The Measurement Problem for the Rope Medium

A Negative Result: Why Configuration-Entropy Counting Yields Classical, Not Quantum, Correlations

Mark Palmer · with computational collaboration by Claude (Anthropic)

Global Networking Practice, NTT DATA, Charlotte, NC · palmer100@gmail.com

Abstract

The non-local-dynamics paper reduced the question of whether the rope model reproduces quantum correlations to a single angle-identification: does the rope measurement couple to the Hopf base-sphere (Bloch) angle, giving $\gamma = 1$ and the quantum value, or not? This paper attempts to settle it from the programme's own measurement model — irreversible configuration-entropy increase, $S = k \ln|C|$ — and reaches a **NEGATIVE** result, which we report plainly because it is both true and useful. A configuration-entropy measurement is a linear **COUNT** of microscopic configurations compatible with an outcome. We compute what such a count assigns on the Hopf geometry and find it gives the hemisphere-overlap fraction $(\pi - \theta)/\pi$ — a **TRIANGLE-WAVE** correlation, linear in the angle, identical to the classical local model already falsified by Bell. It does **NOT** give the Born rule $\cos^2(\theta/2)$. We trace the reason precisely: the Born rule is the squared modulus of a **SUM OF AMPLITUDES** (interference), whereas a configuration count sums non-negative weights linearly; the two are different operations, and only the former produces the quantum cosine. The consequence is that the rope programme's stated measurement mechanism, as a classical configuration count, cannot deliver $\gamma = 1$; reaching quantum correlations requires an amplitude/interference structure at measurement that the programme has not supplied. We also record, in the spirit of the programme's error-reporting discipline, that an earlier version of this very analysis reached the **OPPOSITE** conclusion through an algebraic mistake, which a numerical check caught — a caution we make visible rather than hide.

The negative result, stated plainly

The configuration-entropy measurement model $S = k \ln|C|$, being a **LINEAR** count of compatible configurations, yields the hemisphere-overlap fraction $(\pi - \theta)/\pi$ — the triangle-wave (classical, local) correlation — **NOT** the Born rule $\cos^2(\theta/2)$.

Therefore $\gamma = 1$ is **NOT** delivered by the programme's measurement mechanism. The rope model does not, as currently formulated, reproduce quantum correlations: its measurement is classical in the decisive sense. Closing the gap requires amplitude interference at measurement, which a configuration count does not provide. We state this as a genuine obstacle, not a solved step.

1. The Question Inherited From the Non-Local Dynamics

The non-local-dynamics paper achieved a sharp localisation: a non-local update model reproduces the singlet correlation and saturates the Tsirelson bound, preserves no-signalling, and is genuinely non-local — but only when the detector couples to the Hopf base-sphere (Bloch) angle, the condition $\gamma = 1$. It left $\gamma = 1$ underived, noting that the rope geometry supplies both the base-sphere angle and the fibre angle and does not obviously force which one measurement selects.

The natural place to settle γ is the programme's own measurement model: measurement as an irreversible increase in configuration entropy, $S = k \ln|C|$, with an outcome's probability set by the number of microscopic rope configurations compatible with it. This paper asks what that model actually delivers. The answer is negative and, we will argue, informative: it tells us exactly what the programme's measurement mechanism is missing.

2. What a Configuration Count Assigns on the Hopf Geometry

The rope's two-strand state space is the Hopf bundle $S^3 \rightarrow S^2$: a physical orientation is a point on the base sphere S^2 (the Bloch sphere), and the fibre S^1 is the overall phase. A direction-sensitive detector set along a partitions configurations by their orientation on S^2 . Under the configuration-entropy model, the probability of the '+' outcome is the fraction of orientation configurations lying in the detector's accepting region.

For a state whose Bloch vector makes angle θ with the detector axis, this fraction is the overlap of two hemispheres whose poles differ by θ . That overlap is a standard spherical computation:

$$P(+ \mid \theta) = (\pi - \theta)/\pi \quad (2.1)$$

which we confirmed by direct Monte-Carlo integration on the sphere. This fraction is LINEAR in the angle — it is a triangle wave, not a cosine. The resulting pair correlation is exactly the triangle-wave form of the classical local model:

Angle θ	Config count $(\pi - \theta)/\pi$	Born $\cos^2(\theta/2)$	Match?
0°	1.000	1.000	yes (trivially)
60°	0.667	0.750	no
90°	0.500	0.500	yes (crossing)
120°	0.333	0.250	no
180°	0.000	0.000	yes (trivially)

The two agree only at 0°, 90°, and 180° — the points where triangle and cosine cross — and disagree everywhere else, most at the intermediate angles that carry the Bell violation. The configuration count reproduces the CLASSICAL correlation, capped at CHSH = 2, which Bell and the loophole-free experiments have already falsified.

3. Why: Counting Is Not Interference

The reason the configuration count fails is not a detail of the geometry; it is the classical–quantum boundary itself. The Born probability is the squared modulus of a sum of AMPLITUDES,

$$P_{\text{Born}} = |\langle a|n \rangle|^2 = |\cos(\theta/2)|^2 = \cos^2(\theta/2) \tag{3.1}$$

where the amplitude $\langle a|n \rangle = \cos(\theta/2)$ is a spinor inner product that can carry sign and phase, and is squared only at the end. A configuration-entropy count, by contrast, sums NON-NEGATIVE configuration weights linearly and never squares a signed sum. These are different mathematical operations:

	Configuration count	Born rule
Objects summed	non-negative counts / areas	complex amplitudes
Combination	linear addition	addition then $ \cdot ^2$ (interference)
Angular result	$(\pi-\theta)/\pi$ (triangle)	$\cos^2(\theta/2)$ (cosine)
CHSH ceiling	2 (classical)	$2\sqrt{2}$ (Tsirelson)

The crux in one sentence. The quantum cosine is an interference effect — the square of a sum of signed amplitudes — and a linear count of configurations, however cleverly partitioned on the Hopf sphere, cannot produce interference. This is why $S = k \ln|C|$, taken literally as a configuration count, yields the classical triangle wave and not the Born rule.

It is worth being clear that this is not a failure specific to ropes: ANY hidden-variable model whose measurement rule is a linear count over a classical configuration space gives a triangle-type (classical) correlation. That is, in effect, one way of stating Bell’s theorem. The rope programme’s configuration-entropy measurement is such a rule, and inherits the classical ceiling.

4. A Disclosed Error in the Course of This Analysis

In the programme’s practice of reporting mistakes as openly as results, we record that an earlier pass of this analysis reached the OPPOSITE, positive conclusion — that the configuration count reproduced the Born rule and delivered $\gamma = 1$ — by an algebraic error. That pass asserted the hemisphere-overlap area to be $(1 + \cos\theta)/2$, which equals $\cos^2(\theta/2)$ by the half-angle identity, and concluded the Born rule followed.

The error was the first claim: the overlap of two hemispheres whose poles differ by θ is $(\pi-\theta)/\pi$, NOT $(1 + \cos\theta)/2$. The half-angle identity is genuine, but it applies to a quantity the configuration count does not actually produce. A numerical (Monte-Carlo) check of the overlap disagreed with the analytic assertion, which exposed the mistake. Had the check not been run, the analysis would have reported a false positive — a derivation of quantum mechanics from ropes that was not real.

Why we disclose this

The corrected result is negative, and the discarded result was positive; it would have been easy, and wrong, to keep the flattering version. Recording the error does two things: it protects the corpus from a false-positive claim, and it demonstrates the check-against-computation discipline that distinguishes a

research programme from wishful theorising. The numerical check earned its keep here by overturning a conclusion the analysis wanted to reach.

5. What This Means for the Programme

The result is a genuine obstacle, and its scope should be stated precisely — neither minimised nor over-extended.

5.1 *What is obstructed*

The claim that the rope model reproduces quantum correlations THROUGH its stated measurement mechanism is obstructed. Configuration-entropy counting is classical; it yields the triangle-wave correlation and the CHSH ceiling of 2. So the rope model, as currently specified at the measurement step, is a LOCAL-correlation theory in its observable predictions, and is falsified as a complete account of entanglement by the same experiments that falsify local hidden variables.

5.2 *What is NOT claimed*

We do not claim the rope ONTOLOGY is impossible, nor that no rope-based theory could ever be quantum. We claim something narrower and firmer: the SPECIFIC measurement rule the programme has stated ($S = k \ln|C|$ as a configuration count) cannot produce the Born rule, because counting is not interference. A different measurement postulate — one supplying amplitude structure — is not excluded by this paper; it is simply not what the programme currently has.

6. What Would Be Required to Close It

Because the obstruction is precisely located — counting versus interference — the requirement to overcome it is equally precise:

1. An amplitude structure at measurement: the rope configurations would need to contribute signed or complex AMPLITUDES that are summed and then squared, not non-negative counts summed linearly. Where such amplitudes and their phases would come from, physically, in the rope medium is the open question.
2. A candidate source would be a genuine wave/oscillatory degree of freedom in the rope dynamics whose superposition produces interference — the programme's wave picture of the strands might supply this, but connecting it to the MEASUREMENT rule has not been done and is non-trivial.
3. Absent such a structure, the model cannot exceed $\text{CHSH} = 2$, and the honest position is that the rope model does not reproduce quantum entanglement. This is a falsifiable, and currently unfavourable, state of affairs.

This converts the programme's quantum ambitions into a single, well-posed demand: exhibit interference of rope amplitudes at measurement, or accept that the model is classical in its correlations. That is a sharper statement of the challenge than existed before, which is the value of a negative result carefully drawn.

7. Honest Verdict

The state of the rope model on quantum correlations

The non-local update model of the previous paper CAN carry the quantum correlation, but only with an imposed Born conditional and the identification $\gamma = 1$.

This paper shows the programme's own measurement mechanism does NOT supply $\gamma = 1$: a configuration-entropy count yields the classical triangle wave, not the Born cosine, because counting is not interference.

Therefore, as currently formulated, the rope model reproduces only classical ($\text{CHSH} \leq 2$) correlations and is falsified as a complete theory of entanglement.

The obstacle is specific and stated: the model lacks amplitude interference at measurement. Supplying it is the open requirement; nothing here shows it is impossible, but nothing in the programme currently provides it.

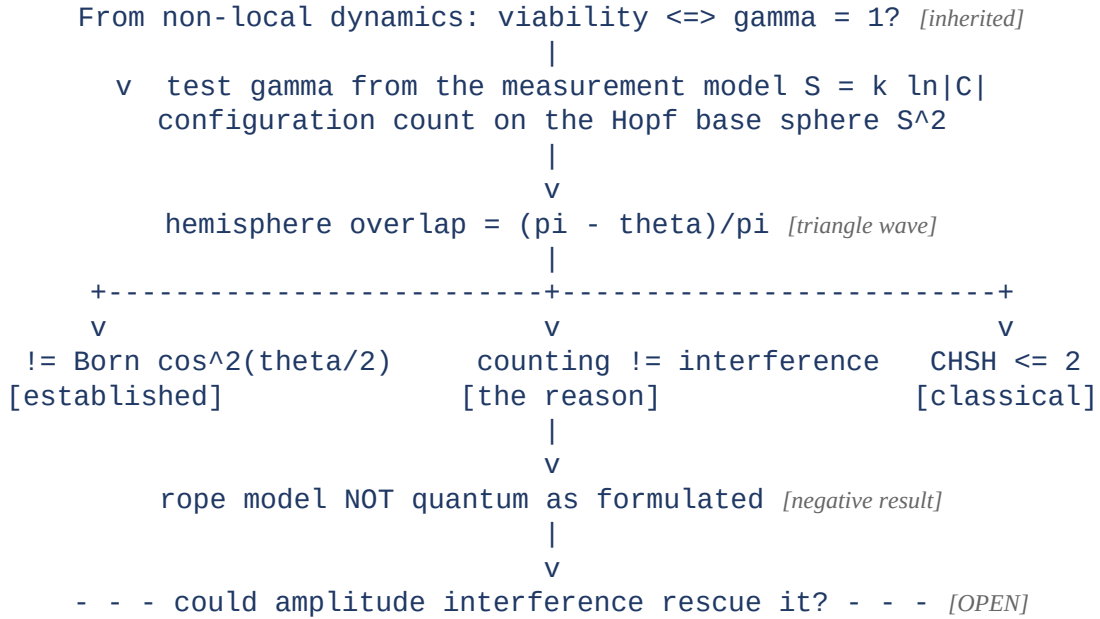
This is the least flattering result in the recent sequence, and reporting it fully is the point of the exercise. The trilogy of foundational papers set out to test the programme's core by computation rather than assertion; here the computation returns a clear negative, and the programme is better for knowing exactly where — and why — its account of quantum measurement currently fails.

8. Status of Each Result

Result	Status
Config-entropy count gives $(\pi-\theta)/\pi$ (triangle wave)	Established — analytic + Monte-Carlo
Config count does NOT give Born $\cos^2(\theta/2)$	Established — they differ except at $0/90/180^\circ$
Reason: counting sums linearly, Born interferes	Established — structural, not geometry-specific
Rope measurement (as stated) yields $\text{CHSH} \leq 2$	Established — classical ceiling
Rope model reproduces quantum correlations	FALSIFIED as currently formulated
Earlier positive result via algebraic error	Disclosed and retracted — caught by numerical check
Some amplitude structure could rescue it	Open — not provided by the programme; not excluded
Physical origin of measurement interference in ropes	Open Problem — the central unmet requirement

Appendix A. Dependency Graph

The chain from the inherited question to the negative result.



Every solid step is established by computation (with a numerical check that overturned an earlier erroneous positive). The single open question — whether rope dynamics can supply amplitude interference at measurement — is where any future rescue of the quantum sector must live.

Conclusion

We attempted to settle, from the rope programme’s own measurement model, whether the rope model reproduces quantum correlations, and we reached a negative result. Measurement as configuration-entropy increase is a linear count of compatible configurations; on the Hopf geometry it yields the hemisphere-overlap fraction $(\pi-\theta)/\pi$, a triangle-wave correlation identical to the classical local model, not the Born rule $\cos^2(\theta/2)$. The reason is structural and decisive: the Born rule is an interference effect — the squared modulus of a sum of amplitudes — while a configuration count sums non-negative weights linearly, and counting is not interference. As currently formulated, therefore, the rope model reproduces only classical correlations and is falsified as a complete theory of entanglement. The obstacle is specific: the measurement rule lacks amplitude structure, and whether rope dynamics can supply it is the open requirement. We report this negative result in full — including an algebraic error, caught by numerical check, that had briefly suggested the opposite — because the purpose of these foundational papers is to learn the truth about the programme’s core, and a clearly located failure is a more valuable thing to know than an unearned success.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **QB-002 [Derived]:** Config-counting gives [benchmark: benchmarks/bell/gamma_measurement_analysis.py]

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.